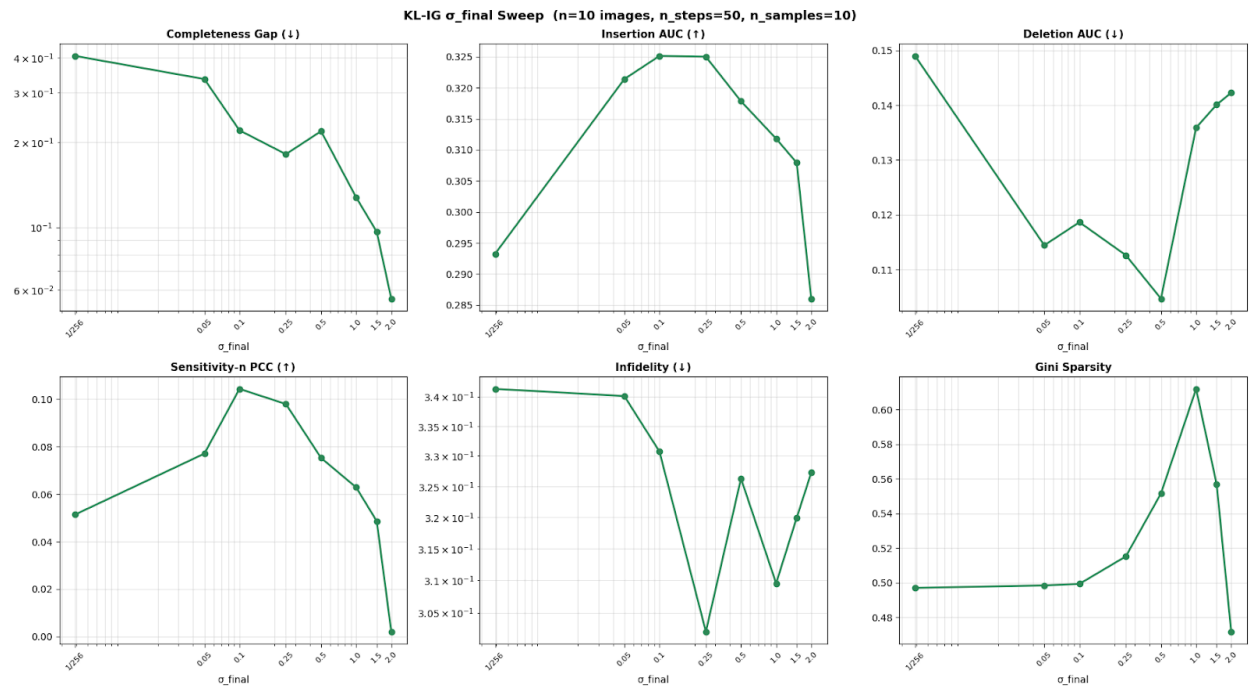
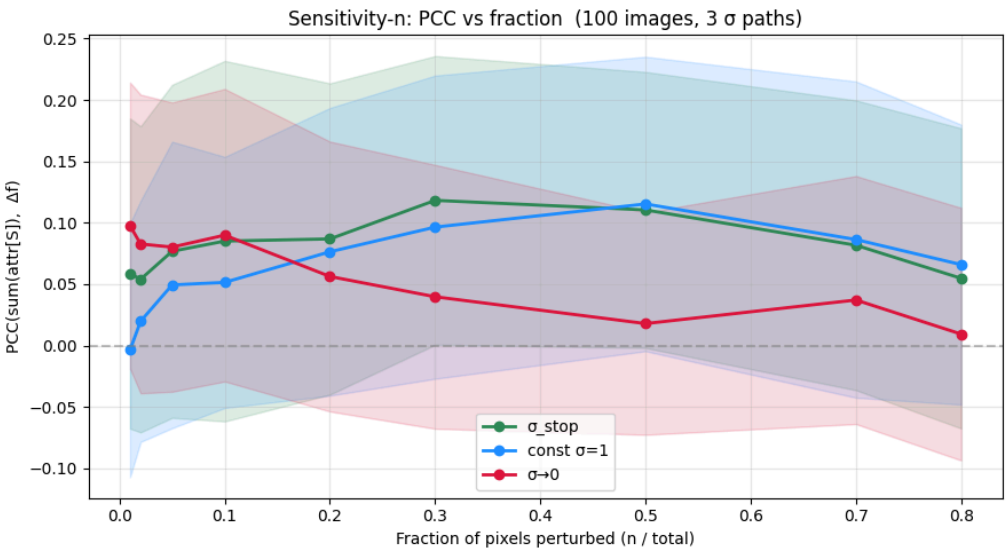
WK3

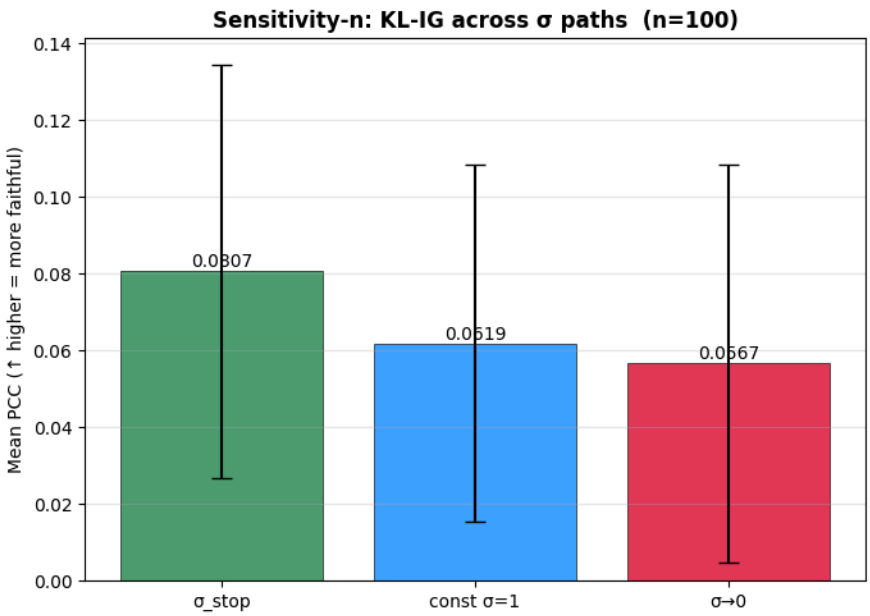
- ## Sigma constant Comparison

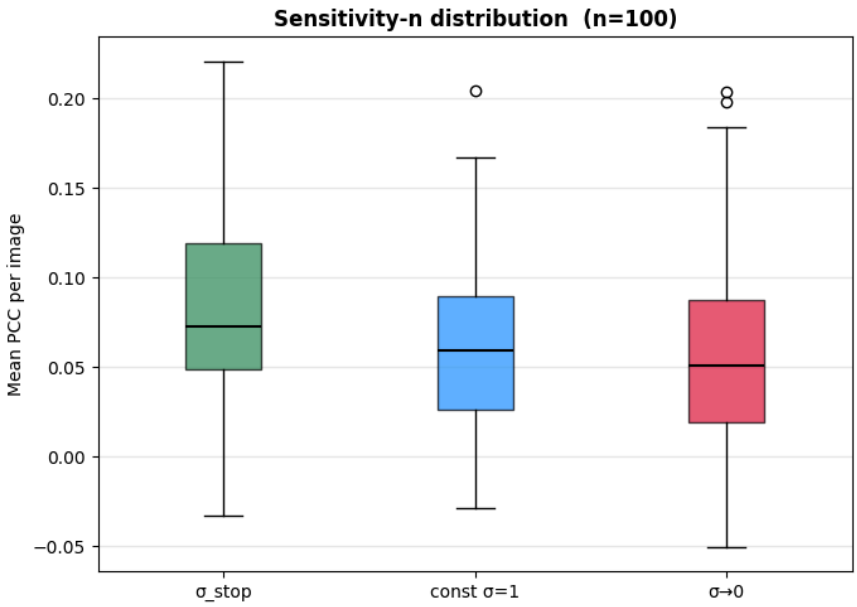


When swept over various constant sigma values, (sigma 0.25 yields optimal performance in most of the metrics)

- **<u>Sigma const, 0 vs sigma stop</u>**

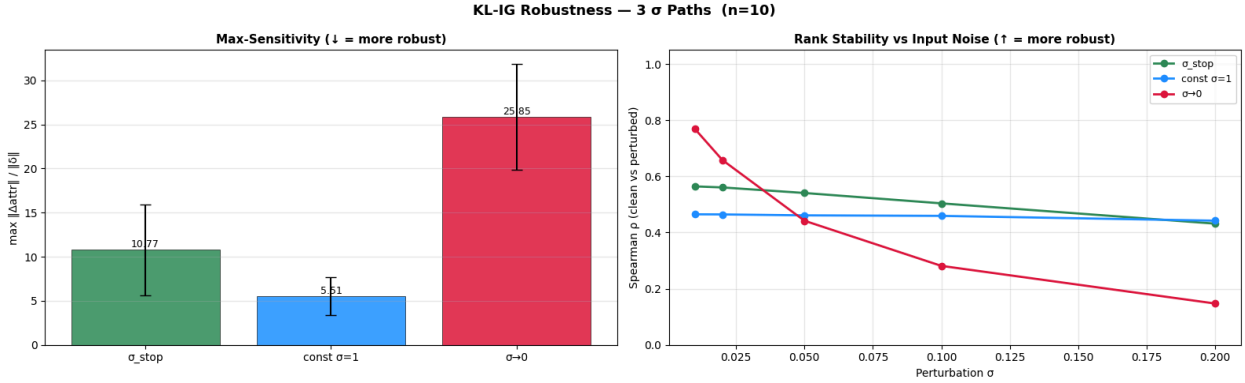






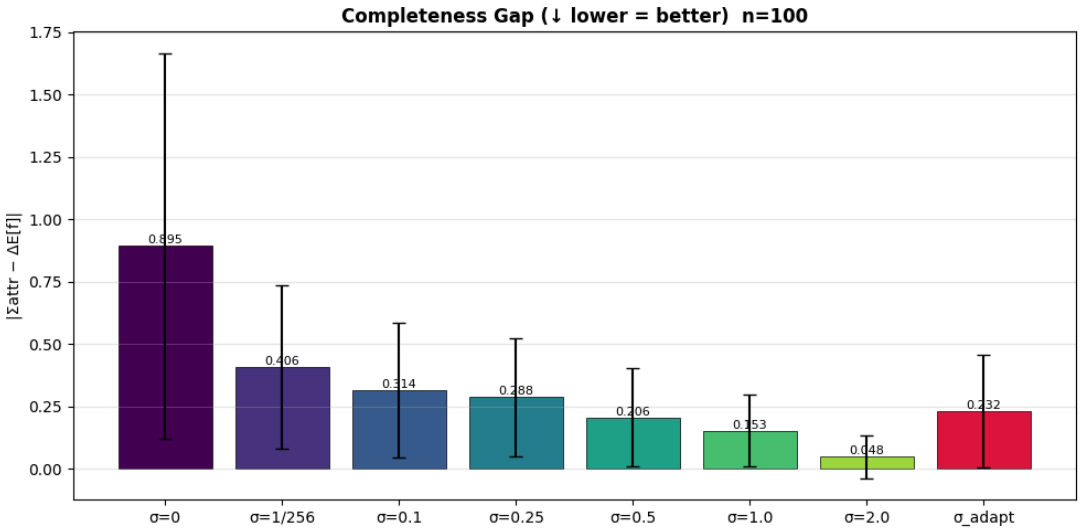
path	mean PCC	std	median
σ_stop	+0.0807	0.0538	+0.0728
const σ=1	+0.0619	0.0465	+0.0595
σ→0	+0.0567	0.0518	+0.0508

2.Robustness

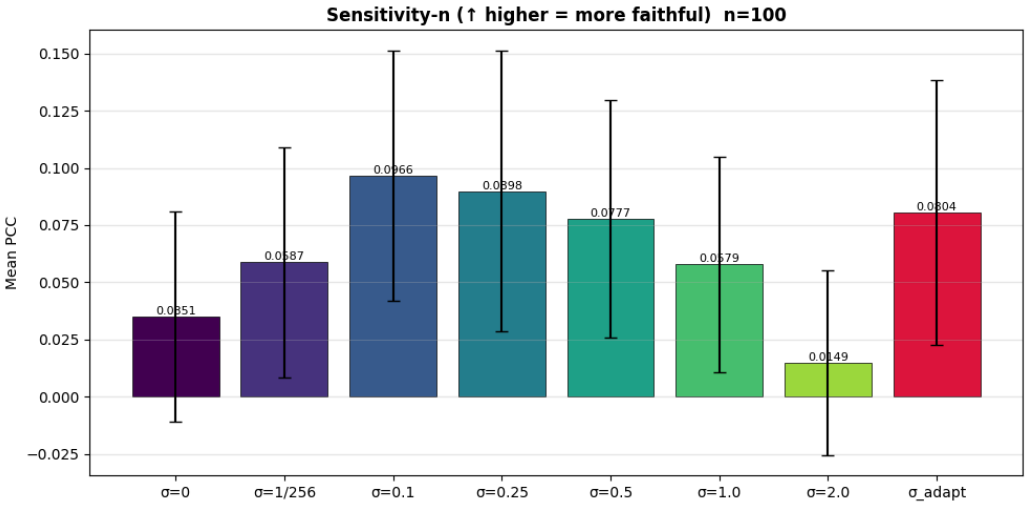


Comprehensive comparison

Completeness gap:

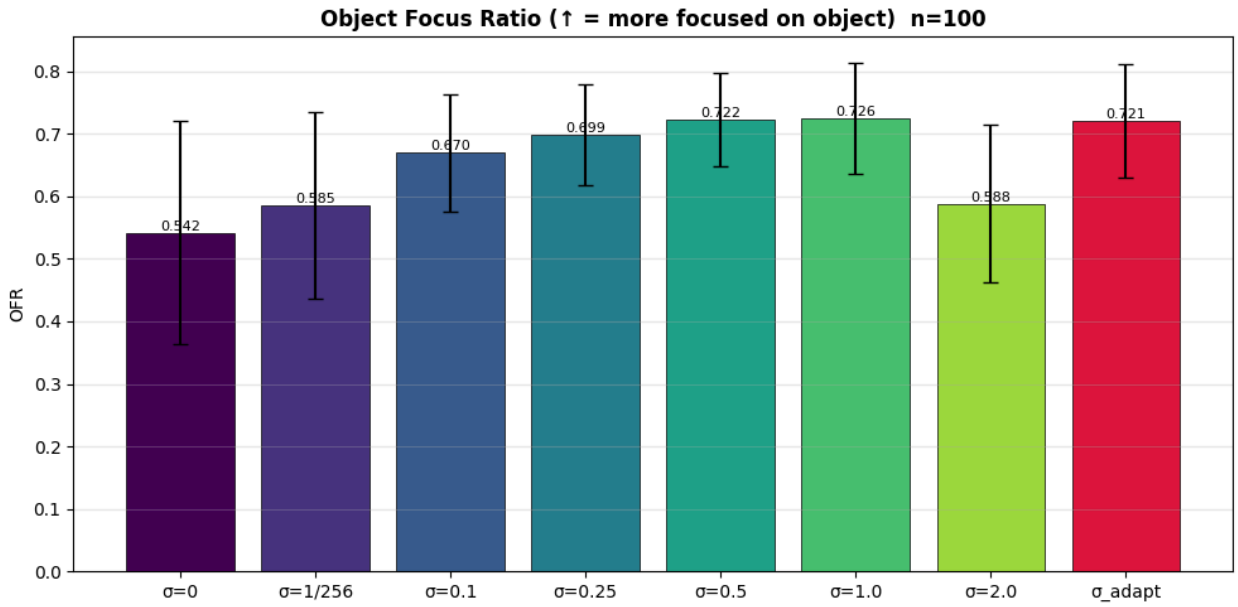


1. Sensitivity-n

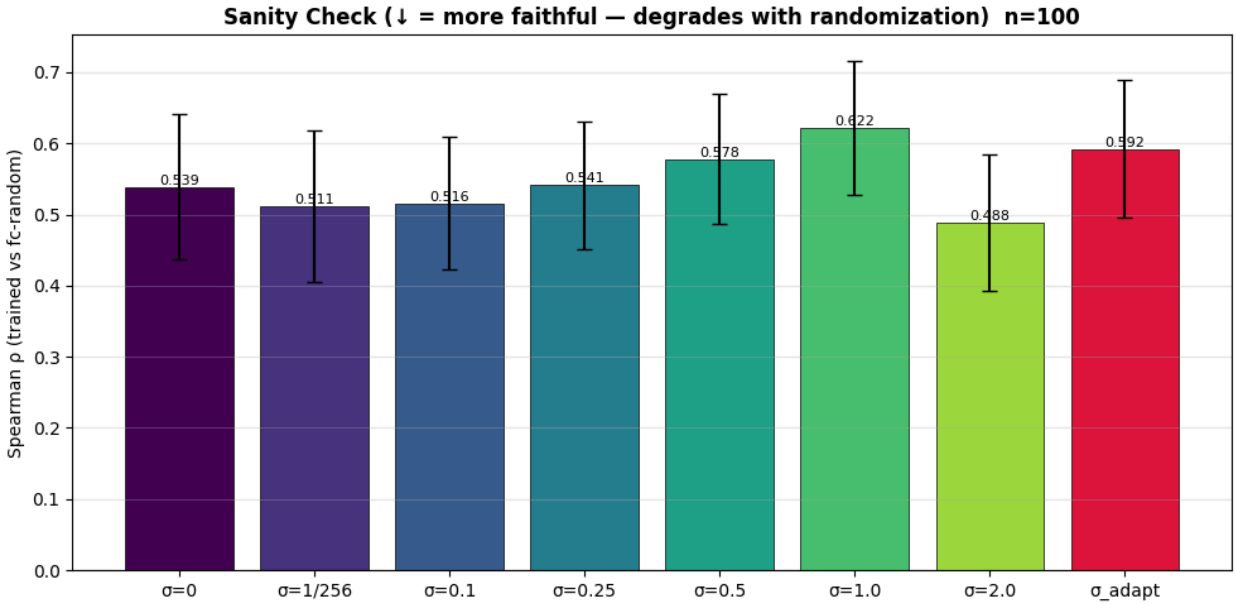


2.

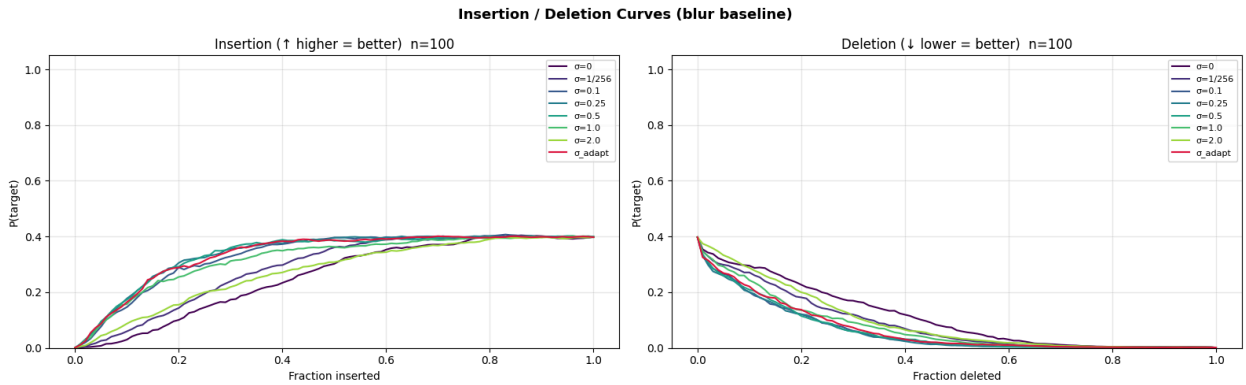
3. OFR



3. Sanity Check



4.



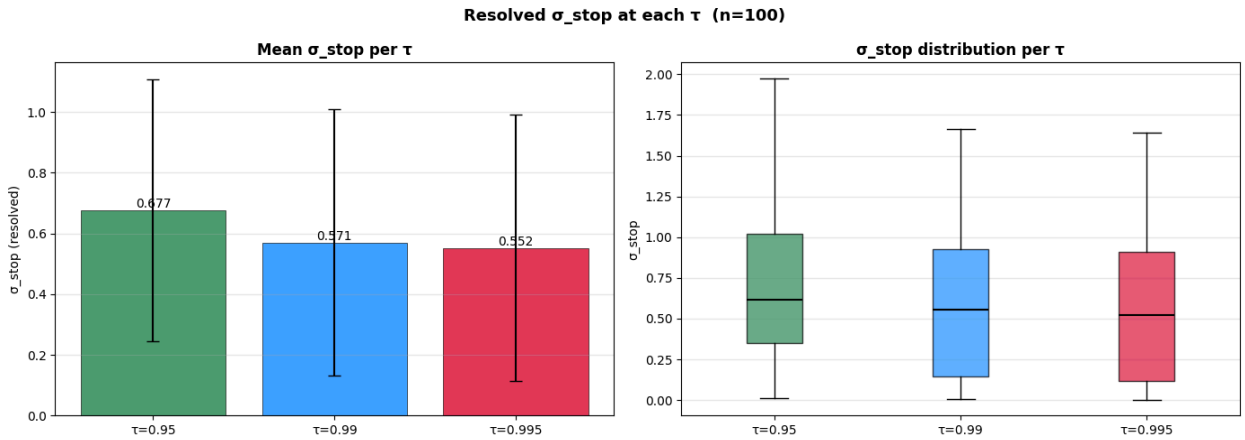
σ_final	σ used	Gap↓	Sens↑	Ins↑	Del↓	OFR↑	Sanity↓
σ=0	0.000	0.8947	0.0351	0.257	0.110	0.542	0.539
σ=1/256	0.004	0.4062	0.0587	0.289	0.086	0.585	0.511
σ=0.1	0.100	0.3137	0.0966	0.332	0.060	0.670	0.516
σ=0.25	0.250	0.2876	0.0898	0.341	0.058	0.699	0.541
σ=0.5	0.500	0.2061	0.0777	0.340	0.061	0.722	0.578
σ=1.0	1.000	0.1529	0.0579	0.321	0.074	0.726	0.622
σ=2.0	2.000	0.0485	0.0149	0.275	0.093	0.588	0.488
σ_adapt	0.671	0.2317	0.0804	0.338	0.065	0.721	0.592

σ=0 and σ=2.0 perform the worst. σ=0 has terrible completeness (0.89 gap) and worst insertion/deletion; σ=2.0 destroys sensitivity-n (0.015) and insertion quality.

σ_adapt is decent but not optimal

Trade off: Completeness gap wants high σ, but (sens-n, ins/del) want moderate σ.

σ=0.25 wins the most metrics outright (ins, del, sens-n runner-up)



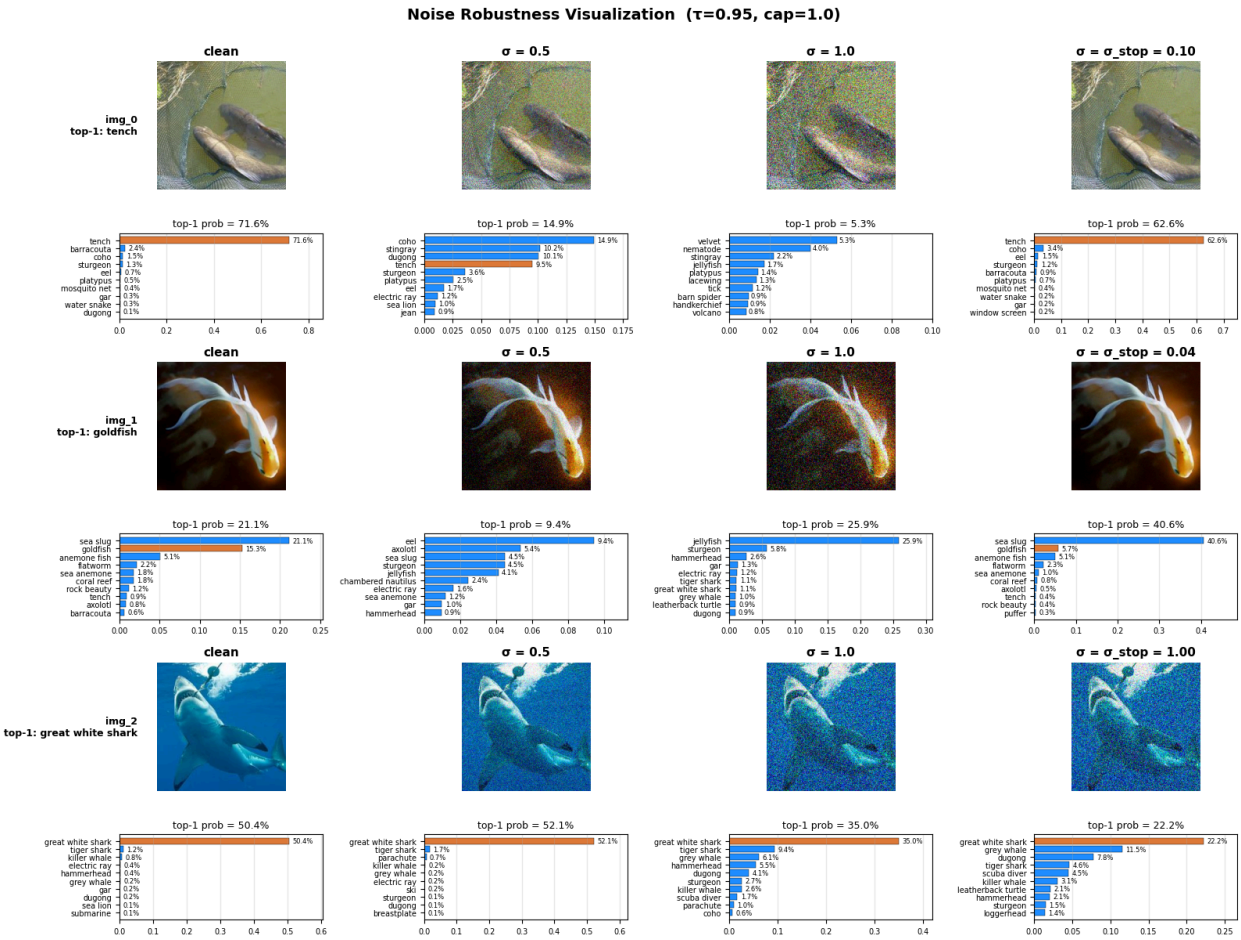
 # Experiment: constrain the tau value: if we tighten τ, does σ_stop at a preferable value

Assumption: Tighter τ → smaller σ_stop

τ	mean σ	std σ	min σ	max σ
0.950	0.677	0.431	0.011	1.973
0.990	0.571	0.439	0.004	1.662
0.995	0.552	0.439	0.004	1.643

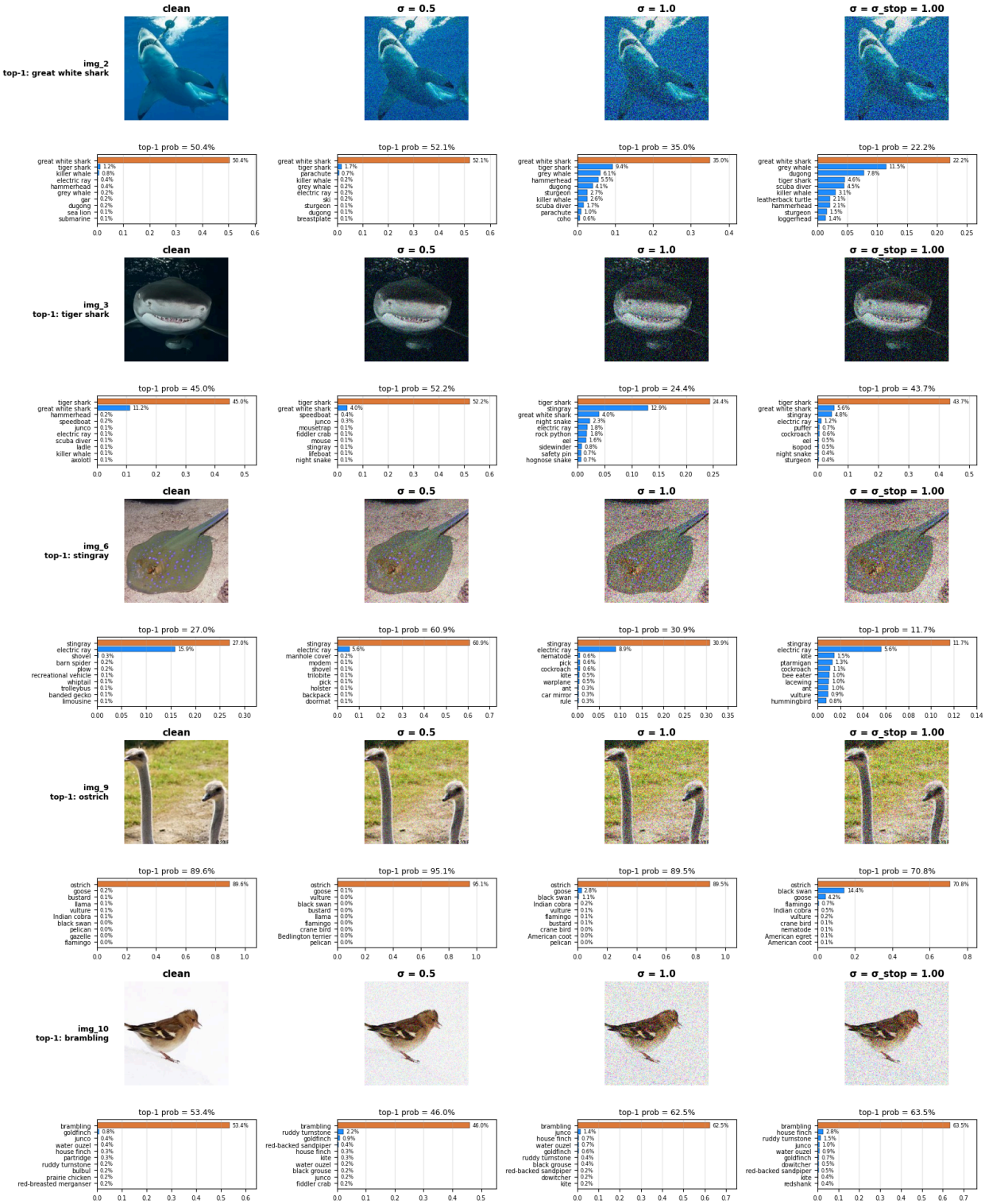
Even at tau=0.995, max σ = 1.64 ; meaning some images still absorb enormous noise

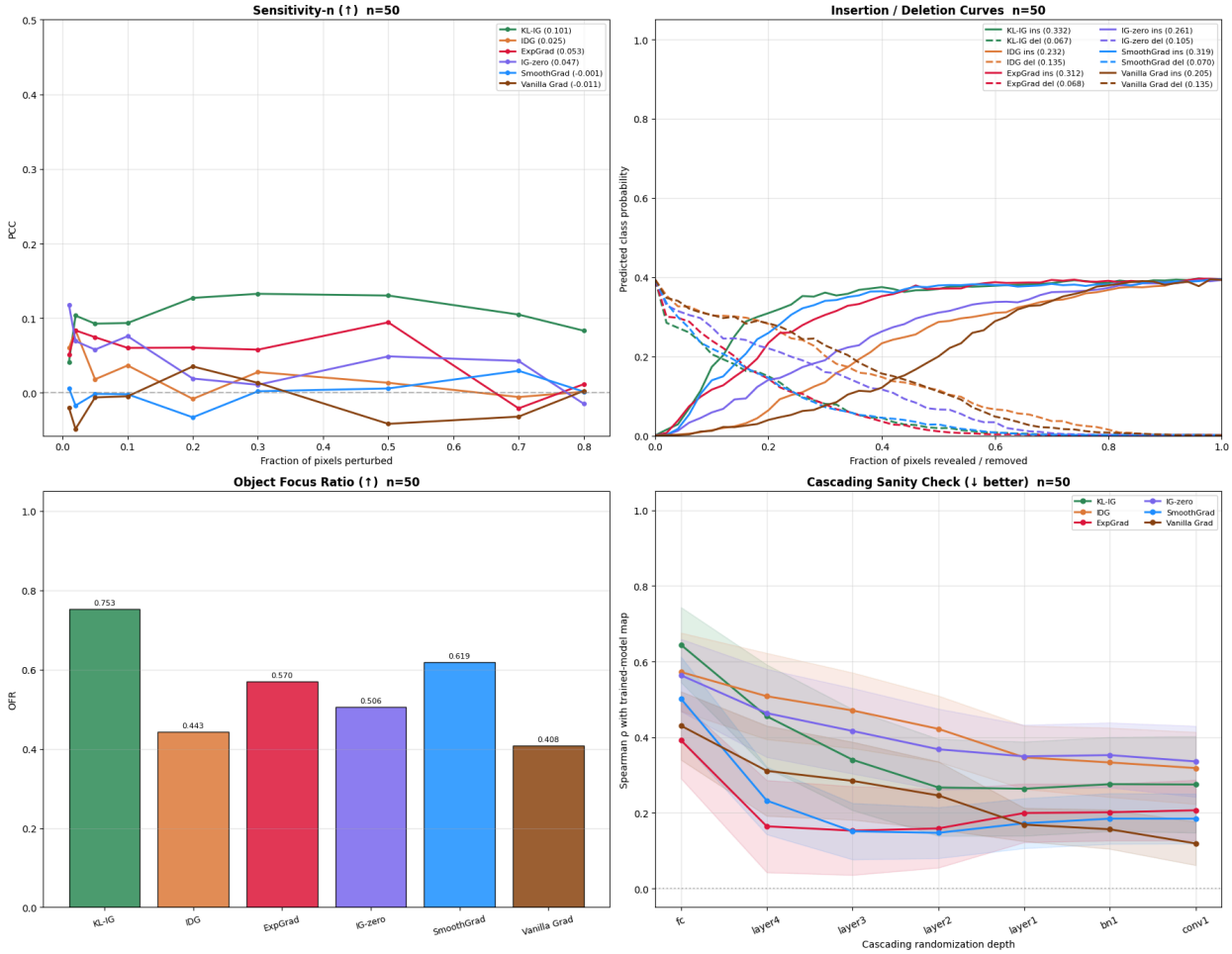
#Experiment: Keep sigma adaptive; but don't let the sigma stop exceeding 1

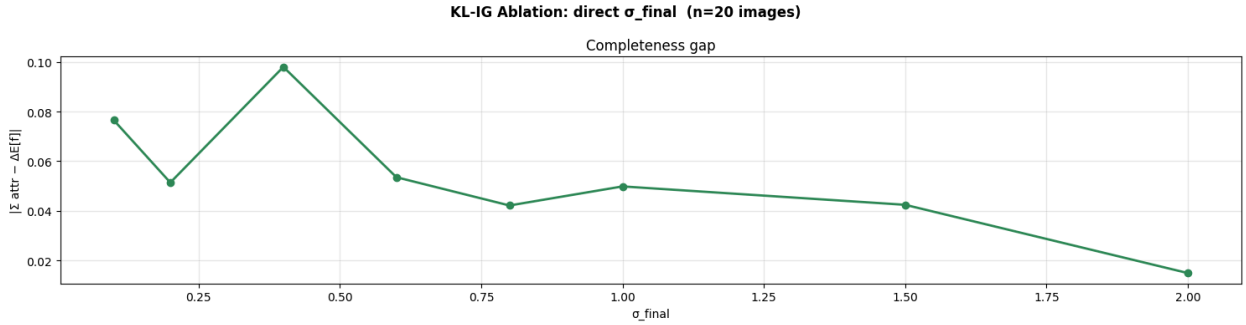


Images with max sigma_stop:

Noise Robustness Visualization — highest σ_stop images (τ=0.95, cap=1.0)




Sensitivity-n (↑) n=50
PCC
Fraction of pixels perturbed
KL-IG (0.101)
IDG (0.025)
ExpGrad (0.053)
IG-zero (0.047)
SmoothGrad (-0.001)
Vanilla Grad (-0.011)
Insertion / Deletion Curves n=50
Predicted class probability
Fraction of pixels revealed / removed
KL-IG ins (0.332)
KL-IG del (0.067)
IDG ins (0.232)
IDG del (0.135)
ExpGrad ins (0.312)
ExpGrad del (0.068)
IG-zero ins (0.261)
IG-zero del (0.105)
SmoothGrad ins (0.319)
SmoothGrad del (0.070)
Vanilla Grad ins (0.205)
Vanilla Grad del (0.135)
Object Focus Ratio (↑) n=50
OFR
0.753
0.443
0.570
0.506
0.619
0.408
KL-IG
IDG
ExpGrad
IG-zero
SmoothGrad
Vanilla Grad
Cascading Sanity Check (↓ better) n=50
Spearman ρ with trained-model map
Cascading randomization depth
fc
layer4
layer3
layer2
layer1
bn1
conv1


KL-IG Ablation: direct σ_final (n=20 images)
Completeness gap
|Σ attr − ΔE[f]|
σ_final